

# HireFlow AI

Next-Generation AI-Powered Applicant Tracking & Talent Evaluation Platform

## System Use Case Specification & Architecture Documentation

|                   |                            |                  |                                        |
|-------------------|----------------------------|------------------|----------------------------------------|
| Document Version: | 1.0.0 (Production Release) | Target Platform: | AWS Amplify (Frontend) + Render (API)  |
| Author / Lead:    | Thejas                     | Database:        | MongoDB Atlas Cloud Cluster            |
| Date of Release:  | August 2026                | AI Engine:       | OpenAI GPT-4o-mini + Word-Boundary NLP |

## 1. Executive Summary & Solution Overview

HireFlow AI is a full-stack, enterprise-grade Talent Operating System and Applicant Tracking System (ATS) engineered to bridge candidate discovery with autonomous AI-driven candidate evaluation. It replaces fragmented hiring spreadsheets with automated resume decoding, authenticity/plagiarism detection, strict forward-only hiring stage governance, and cloud-native PDF persistence.

| Layer               | Technology                                 | Key Responsibilities                                                                |
|---------------------|--------------------------------------------|-------------------------------------------------------------------------------------|
| Frontend UI/UX      | React 18, Vite, Tailwind CSS, Lucide Icons | Responsive public portal, dedicated mobile candidate cards, admin command center.   |
| Cloud Hosting       | AWS Amplify (Web) + Render (Node)          | Automated CI/CD git-triggered builds, SSL termination, and global CDN delivery.     |
| Backend Server      | Node.js, Express, Multer, JWT, CORS        | REST API endpoints, JWT security guard, resume binary streaming, stage transitions. |
| Database Layer      | MongoDB Atlas (Mongoose ODM)               | Cloud Base64 PDF storage, candidate indexing, interview logs, duplicate protection. |
| Intelligence Engine | OpenAI GPT-4o-mini & NLP Heuristic         | Skill extraction, role match scoring (0-100%), plagiarism risk & originality audit. |

## 2. Primary System Actors

| Actor                     | Description & Operational Scope                                                                                                                                                                        |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Job Applicant (Candidate) | Public user browsing active positions, submitting PDF resumes with personal credentials, and receiving fast-track application confirmations.                                                           |
| Recruiter / Hiring Team   | Authenticated admin managing requisitions, inspecting AI candidate scoring, viewing streamed PDF resumes, conducting multi-round interview ratings, and advancing candidates along strict stage gates. |
| AI Intelligence Service   | Autonomous background engine that extracts keywords, assesses technical depth, calculates originality vs plagiarism risk, and compiles strengths and gaps.                                             |

### 3. Detailed Use Case Specifications

#### UC-01: Job Requisition Discovery & Search

|                                 |                                                                                                                                                                                                                                                                                                                                             |
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| <b>Primary Actor:</b>           | Job Applicant / Public Visitor                                                                                                                                                                                                                                                                                                              |
| <b>Preconditions:</b>           | Public access to HireFlow AI landing page.                                                                                                                                                                                                                                                                                                  |
| <b>Trigger:</b>                 | Candidate visits home URL or clicks 'Open Positions'.                                                                                                                                                                                                                                                                                       |
| <b>Main Success Flow:</b>       | <ol style="list-style-type: none"><li>1. System loads active job listings from database via GET /api/jobs.</li><li>2. Candidate searches by role keywords, skills, or location.</li><li>3. Candidate filters by employment type (Full-Time, Contract, Internship).</li><li>4. Candidate clicks 'Apply Now' on target requisition.</li></ol> |
| <b>Exceptions / Alternates:</b> | 4a. If no jobs match search criteria, system presents helpful empty state with reset filters button.                                                                                                                                                                                                                                        |
| <b>Postconditions:</b>          | Candidate is navigated to Apply page with target jobId pre-selected.                                                                                                                                                                                                                                                                        |

#### UC-02: Application Submission & Atlas PDF Storage

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>           | Job Applicant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Preconditions:</b>           | Target job is active and accepting applications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Trigger:</b>                 | Candidate submits application form with PDF resume upload.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Main Success Flow:</b>       | <ol style="list-style-type: none"><li>1. Candidate inputs Full Name, Email, Phone, LinkedIn, and optional cover note.</li><li>2. Candidate uploads PDF resume file (max 5MB).</li><li>3. Backend parses multipart form data via Multer and converts PDF buffer to Base64 string.</li><li>4. System saves Candidate document and Application record in MongoDB Atlas.</li><li>5. System initializes hiring stage at <b>Applied</b> and triggers asynchronous AI parsing.</li><li>6. UI redirects candidate to Application Success page with summary details.</li></ol> |
| <b>Exceptions / Alternates:</b> | <p>3a. Duplicate email applied to same job: System returns 409 Conflict with clear error message.</p> <p>3b. Invalid file format / size &gt; 5MB: System rejects upload with 400 Bad Request.</p>                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Postconditions:</b>          | Application and PDF resume are permanently persisted in MongoDB Atlas cloud.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

#### UC-03: AI Resume Extraction & Skill Alignment

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| <b>Primary Actor:</b>           | AI Intelligence Engine / Recruiter                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Preconditions:</b>           | Application submitted with PDF resume data.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Trigger:</b>                 | Automated on application creation or triggered via 'Re-run Analysis'.                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Main Success Flow:</b>       | <ol style="list-style-type: none"><li>1. AI Service decodes Base64 resume buffer and extracts text.</li><li>2. OpenAI GPT-4o-mini prompt analyzes resume against target job description.</li><li>3. System extracts candidate skills list and determines matched vs missing competencies.</li><li>4. Computes Match Score (0-100%), verified strengths list, and potential interview gap probes.</li><li>5. Results saved to application document under aiAnalysis schema.</li></ol> |
| <b>Exceptions / Alternates:</b> | 2a. OpenAI API key unavailable or rate limited: System seamlessly falls back to local NLP heuristic word-boundary parser without user disruption.                                                                                                                                                                                                                                                                                                                                    |
| <b>Postconditions:</b>          | Candidate profile displays accurate Match Score radial ring and extracted skills breakdown.                                                                                                                                                                                                                                                                                                                                                                                          |

#### UC-04: Resume Plagiarism & Authenticity Audit

|                       |                                                 |
|-----------------------|-------------------------------------------------|
| <b>Primary Actor:</b> | AI Intelligence Engine / Recruiter              |
| <b>Preconditions:</b> | Resume text extracted and ready for evaluation. |

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| <b>Trigger:</b>                 | Triggered as part of AI Candidate Insights pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Main Success Flow:</b>       | <ol style="list-style-type: none"><li>1. System scans resume text for boilerplate buzzword density, template clichés, and project specificity.</li><li>2. Computes <b>Originality Score (%)</b> and <b>Plagiarism Risk (%)</b>.</li><li>3. Assigns risk categorization: Low Risk (&lt;=20%), Standard Phrasing (21-40%), High Template Plagiarism (&gt;40%).</li><li>4. Renders dual-color progress gauge and authenticity observation flag on recruiter card.</li></ol> |
| <b>Exceptions / Alternates:</b> | 1a. Short or non-technical resume: System flags lack of quantified metrics and raises alert.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Postconditions:</b>          | Recruiter is empowered with objective authenticity rating to screen out copied resumes.                                                                                                                                                                                                                                                                                                                                                                                  |

#### UC-05: Strict Forward-Only Stage Governance

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| <b>Primary Actor:</b>           | Recruiter / Hiring Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Preconditions:</b>           | Authenticated admin viewing candidate details at /admin/applications/:id.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Trigger:</b>                 | Recruiter clicks 'Update Stage' button.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Main Success Flow:</b>       | <ol style="list-style-type: none"><li>1. System opens Stage Decision Modal displaying current stage (e.g. Applied).</li><li>2. Modal presents two forward paths: ■ <b>Selected / Advance</b> (→ R1) OR ■ <b>Rejected</b> (→ Reject).</li><li>3. Recruiter enters mandatory decision rationale note.</li><li>4. Backend validates transition against strict forward state machine:<ul style="list-style-type: none"><li>• Applied → R1   Reject</li><li>• R1 → R2   R1 Reject</li><li>• R2 → R3   R2 Reject</li><li>• R3 → Approved   R3 Reject</li></ul></li><li>5. Stage history log updated with timestamp and decision note.</li></ol> |
| <b>Exceptions / Alternates:</b> | <ol style="list-style-type: none"><li>4a. Recruiter attempts illegal backward move: System rejects transition with validation error.</li><li>4b. Candidate already in terminal stage (Approved / Reject): Modal locks stage selector.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Postconditions:</b>          | Candidate stage updated and reflected across dashboard, timeline, and mobile cards.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

#### UC-06: Inline Cloud PDF Resume Streaming

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor:</b>           | Recruiter / Hiring Manager                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Preconditions:</b>           | Candidate document contains Base64 resume buffer in Atlas.                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Trigger:</b>                 | Recruiter clicks 'View Resume (PDF)' in Admin portal.                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Main Success Flow:</b>       | <ol style="list-style-type: none"><li>1. Request dispatched to GET /api/applications/:id/resume.</li><li>2. Backend retrieves candidate.resumeData from MongoDB Atlas.</li><li>3. Decodes Base64 buffer to binary Buffer.from(data, 'base64').</li><li>4. Sets response headers: Content-Type: application/pdf, Content-Disposition: inline.</li><li>5. Browser opens resume PDF in a dedicated tab with native PDF controls (zoom, print, save).</li></ol> |
| <b>Exceptions / Alternates:</b> | 2a. Resume data missing / legacy upload: Backend attempts local file fallback or returns 404.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Postconditions:</b>          | Recruiter inspects original candidate PDF with zero dependency on ephemeral cloud storage.                                                                                                                                                                                                                                                                                                                                                                  |

#### UC-07: Structured Interview Feedback & Ratings

|                       |                                                                      |
|-----------------------|----------------------------------------------------------------------|
| <b>Primary Actor:</b> | Recruiter / Technical Interviewer                                    |
| <b>Preconditions:</b> | Candidate is in active interview round (R1, R2, or R3).              |
| <b>Trigger:</b>       | Interviewer logs feedback in 'Interview Notes & Evaluation' section. |

|                                 |                                                                                                                                                                                                                                                                                                                                   |
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| <b>Main Success Flow:</b>       | 1. Recruiter selects Interview Round (R1 Technical, R2 Deep Tech, R3 System Design, HR).<br>2. Assigns 1 to 5 Star Rating.<br>3. Enters detailed technical observations, questions asked, and strengths.<br>4. Clicks 'Save Interview Note'.<br>5. System persists note in candidate interviewNotes array and refreshes timeline. |
| <b>Exceptions / Alternates:</b> | 3a. Empty notes field: System validates input and displays inline prompt.                                                                                                                                                                                                                                                         |
| <b>Postconditions:</b>          | Feedback is permanently chronologically indexed for collaborative hiring decisions.                                                                                                                                                                                                                                               |

4. System Governance, Security & Quality Matrix

| Governance Dimension       | Implementation Mechanism                            | Impact & Guarantee                                                            |
|----------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------|
| Authentication & RBAC      | JWT Bearer Tokens + bcrypt salted password hashing  | Guarantees admin endpoints are shielded from unauthorized access.             |
| Duplicate Guard            | MongoDB compound index on (jobId + candidate email) | Prevents candidates from spamming duplicate applications for same role.       |
| Pipeline Integrity         | Deterministic finite state machine (FSM) validation | Enforces strict forward-only progression; prevents illegal stage regressions. |
| Ephemeral Cloud Resilience | Atlas Base64 direct buffer storage & streaming      | Ensures 100% resume persistence across container restarts on Render/Amplify.  |
| Mobile Responsiveness      | Dedicated mobile cards view (sm:hidden / md:table)  | Flawless touch usability on 360px+ devices without layout clipping.           |

5. Document Authorization & Sign-Off

This Use Case Document certifies that the HireFlow AI application meets all architectural, functional, and governance specifications for production deployment across AWS Amplify and Render cloud environments.

|                     |                             |                     |                      |
|---------------------|-----------------------------|---------------------|----------------------|
| <b>Prepared By:</b> | Thejas (Lead Engineer)      | <b>Project:</b>     | HireFlow AI Platform |
| <b>Status:</b>      | Verified & Production Ready | <b>Approved By:</b> | System Architect     |